

Finite Fourier Analysis

1 The roots of unity filter

The crux of this section turns on one very simple proposition, the so-called *roots of unity filter*:

Proposition 1.1 (Roots of unity filter form 1)

Let ζ be an n -th root of unity and a a positive integer. Then

$$\sum_{j=0}^{n-1} \zeta^{ja} = \begin{cases} n & \text{if } n \mid a, \\ 0 & \text{otherwise.} \end{cases}$$

Proof. If $n \mid a$, then each term in the sum is 1, so things are clear. Otherwise, write

$$\sum_{j=0}^{n-1} \zeta^{ja} = \frac{\zeta^{an} - 1}{\zeta^a - 1} = 0.$$

□

An alternative form concerning generating functions is also commonly used:

Proposition 1.2 (Roots of unity filter form 2)

Consider a generating function

$$F(X) = c_0 + c_1X + c_2X^2 + \cdots$$

Then

$$c_0 + c_n + c_{2n} + \cdots = \frac{f(1) + f(\zeta) + \cdots + f(\zeta^{n-1})}{n}.$$

An oft-cited example of the roots of unity filter are problems of the following flavour.

Example

Compute

$$\binom{2020}{0} + \binom{2020}{4} + \binom{2020}{8} + \cdots + \binom{2020}{2020}.$$

Solution. By [proposition 1.2](#), the answer is

$$\frac{1}{4}(2^{2020} + (1+i)^{2020}(1-i)^{2020} + 0^{2020}) = \frac{1}{4}(2^{2020} + (-4)^{505} + (-4)^{505}) = 2^{2018} - 2^{1008}.$$

□

Equipped with the roots of unity filter, we can transform certain problems of combinatorics into problems of algebra. A more complex (pardon the pun) example showcases the power of this method. It first appeared in competitive mathematics in a Bulgarian TST in 2006 before making a reprisal in the British MO in 2019.

Example (Bulgaria TST 2006, British MO 2019)

Find the number of the subsets B of the set $A = \{1, 2, \dots, p-1\}$ such that the sum of the elements of B is divisible by p .

Solution. Let N denote the subsets with the desired property and ζ a p -th root of unity. (We count the empty subset.) By [proposition 1.1](#),

$$N = \frac{1}{p} \sum_{B \subseteq A} \sum_{j=0}^{p-1} \zeta^{j \sum_{b \in B} b}.$$

Hence the combinatorial problem, which is not at all an easy one, is reduced to a purely algebraic exercise. After swapping the order of summation,

$$\begin{aligned} N &= \frac{1}{p} \sum_{j=0}^{p-1} \sum_{B \subseteq A} \zeta^{j \sum_{b \in B} b} \\ &= \frac{1}{p} \sum_{j=0}^{p-1} \prod_{k=1}^{p-1} (1 + \zeta^{jk}) \end{aligned}$$

It is worth noting that the expressions that result from swapping the order of summation after applying the roots of unity often behave very differently when $j = 0$. Indeed, in this case, when $j = 0$ the inner product evaluates to 2^{p-1} . On the other hand, if $j \neq 0$, then $(\zeta^j, \zeta^{2j}, \dots, \zeta^{(p-1)j})$ is a permutation of $(\zeta, \zeta^2, \dots, \zeta^{p-1})$. Hence

$$N = \frac{1}{p} \left(2^{p-1} + (p-1) \prod_{k=1}^{p-1} (1 + \zeta^k) \right).$$

It is possible to evaluate the final product via direct expansion. We proceed with a more efficient method, however. Let

$$\begin{aligned} f(x) &= 1 + x + \dots + x^{p-1} = \prod_{k=1}^{p-1} (1 - \zeta^k) \\ \implies \prod_{k=1}^{p-1} (1 + \zeta^k) &= (-1)^{p-1} f(-1) = 1 - 1 + \dots + 1 = 1. \end{aligned}$$

Thus $N = \frac{2^{p-1} + p - 1}{p}$.

□

This problem demonstrates a number of important themes about the roots of unity filter.

1. Swapping sums is *extremely* useful and sometimes even the entirety of the problem.
2. After swapping sums, take special care of the case $j = 0$.
3. Solutions will often invoke properties of the roots of unity, particularly their relationship with polynomials.

Notice that it is possible to skip the first few steps in the above solutions if instead [Proposition 1.2](#) is applied to the generating function $f(X) = \prod_{k=1}^{p-1} (1 + X^k)$. The step of swapping the sums and rewriting the expressions is essentially equivalent to finding the generating function for the problem at hand.

Why, then, even bother showing [proposition 1.1](#) at all? There are two reasons. The first is that in some instances it is not easy and also not necessary to spot the relevant generating function. Problems [3.5](#) and [3.15](#) are good examples of this.

Second, [proposition 1.1](#) exemplifies the general usefulness of indicator functions, a profoundly useful tool which is by no means limited to merely the roots of unity filter as seen in olympiad mathematics.

2 Gauss sums

This section assumes a familiarity with the properties of quadratic residues.

Example 2.1 (Online Math Olympiad Fall 2017, Generalisation)

Let p be a prime and n a positive integer. Find the number of solutions to the congruence

$$x_1^2 + x_2^2 + \cdots + x_n^2 \equiv 1 \pmod{p}.$$

Solution. Let N be the number of solutions and $\zeta = e^{\frac{2\pi i}{p}}$. Applying [proposition 1.1](#),

$$\begin{aligned} N &= \frac{1}{p} \sum_{x_i \in \mathbb{F}_p} \sum_{j=0}^{p-1} \zeta^{j(x_1^2 + x_2^2 + \cdots + x_n^2 - 1)} \\ &= \frac{1}{p} \sum_{j=0}^{p-1} \zeta^{-j} \left(\prod_{x=0}^{p-1} \zeta^{jx^2} \right) \\ &= p^{n-1} + \frac{1}{p} \sum_{j=1}^{p-1} \zeta^{-j} \left(\sum_{x=0}^{p-1} \zeta^{jx^2} \right)^n \end{aligned}$$

Let $G(a) = \sum_{x=0}^{p-1} \zeta^{ax^2}$. These are known as the *quadratic Gauss sums modulo p* . For brevity's sake, we also write $G(1) = G$. There are several well known facts about the Gauss sums that are not difficult to prove

Claim 1.

$$G(a) = \left(\frac{a}{p} \right) G$$

Proof. When p divides a , things are clear. If $\left(\frac{a}{p} \right) = 1$, then $x \rightarrow ax^2$ is a permutation of the quadratic residues mod p . Hence, in this case $G(a) = G$.

On the other hand, if $\left(\frac{a}{p} \right) = -1$, then $x \rightarrow ax^2$ maps to zero exactly once and to each quadratic non-residue twice. Hence

$$G(a) = 1 + 2 \sum_{\left(\frac{x}{p} \right) = -1} \zeta^x = 2 \sum_{x=0}^{p-1} \zeta^x - 2 \sum_{\left(\frac{x}{p} \right) = 1} \zeta^x - 1 = -G.$$

Having shown all possible cases, this concludes the proof. $\square$

Thus

$$N = p^{n-1} + \frac{1}{p} \sum_{j=1}^{p-1} \zeta^{-j} \left(\frac{j}{p} \right)^n G^n.$$

If n is even, then all that is left to is compute G . Conversely, if n is odd we will need to invoke the following proposition:

Claim 2.

$$G = \sum_{x=0}^{p-1} \left(\frac{x}{p} \right) \zeta^x$$

Proof. Rewrite

$$\begin{aligned}
 \sum_{x=0}^{p-1} \left(\frac{x}{p}\right) \zeta^x &= \sum_{\left(\frac{x}{p}\right)=1} \zeta^x - \sum_{\left(\frac{x}{p}\right)=-1} \zeta^x \\
 &= 2 \sum_{\left(\frac{x}{p}\right)=1} \zeta^x - \left(\sum_{\left(\frac{x}{p}\right)=1} \zeta^x + \sum_{\left(\frac{x}{p}\right)=-1} \zeta^x \right) \\
 &= G - 1 - (-1) = G.
 \end{aligned}$$

□

Therefore

$$N = \begin{cases} p^{n-1} - \frac{G^n}{p} & \text{if } n \text{ is even,} \\ p^{n-1} - \frac{G^{n+1}}{p} & \text{if } n \text{ is odd.} \end{cases}$$

We don't even need a precise value of G , then. It is enough to find G^2 , which is much easier.

Claim 3.

$$G^2 = p \left(\frac{-1}{p} \right).$$

Proof. By [claim 1](#)

$$\begin{aligned}
 G^2 &= \left(\frac{-1}{p} \right) G \cdot G(-1) \\
 &= \left(\frac{-1}{p} \right) \sum_{x=0}^{p-1} \sum_{y=0}^{p-1} \zeta^{x^2-y^2} \\
 &= \left(\frac{-1}{p} \right) \sum_{x,y} \zeta^{(x-y)(x+y)}.
 \end{aligned}$$

Because $(x, y) \rightarrow (x-y, x+y)$ is a bijection, it follows

$$\sum_{x,y} \zeta^{(x-y)(x+y)} = \sum_{u=0}^{p-1} \sum_{v=0}^{p-1} \zeta^{uv} = p$$

and the claim follows immediately. □

We conclude

$$N = \begin{cases} p^{n-1} - (-1)^{\frac{n(p-1)}{4}} p^{\frac{n}{2}-1} & \text{if } n \text{ is even} \\ p^{n-1} - (-1)^{\frac{(n+1)(p-1)}{4}} p^{\frac{n-1}{2}} & \text{if } n \text{ is odd.} \end{cases}$$

□

A very nice corollary of this is the famous Law of Quadratic Reciprocity.

Corollary (Law of Quadratic Reciprocity)

Let p and q be odd primes. Then

$$\left(\frac{p}{q}\right) \left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2} \cdot \frac{q-1}{2}}.$$

Proof. Take $n = q$ in [example 2.1](#). Any tuple $(x_1, x_2, \dots, x_q)$ can be cyclically shuffled in q ways to produce distinct solutions unless the x_i are all equal. The number of ways for the x_i to be all equal is precisely $\left(\frac{q}{p}\right) + 1$. Thus

$$p^{q-1} + (-1)^{\frac{p-1}{2}, \frac{q-1}{2}} p^{\frac{q-1}{2}} \equiv \left(\frac{q}{p}\right) \pmod{q}.$$

Recall that $p^{\frac{q-1}{2}} \equiv \text{legendrep } q \pmod{q}$, and the result is immediate. □

3 Problems

Many problems are taken from *Problems from the Book* by Titu Andreescu and Gabriel Dospinescu.

Problem 3.1 (IMC 1999). We roll a regular die n times. What is the probability that the sum of the numbers shown is a multiple of 5?

Problem 3.2. Let n be a positive integer and p an odd prime. Find the number of tuples $(x_1, \dots, x_{p-1})$ of positive integers

$$p \mid x_1 + 2x_2 + \dots + (p-1)x_{p-1}$$

and $x_i \leq n$.

Problem 3.3 (IMO Shortlist 1987). Three persons, A, B, C play the following game: A subset with k elements of the set $\{1, 2, \dots, 1986\}$ uniformly at random. The winner is A, B , or C according to whether the sum of the elements of the selected subset is congruent to 0, 1, or 2 modulo 3. Find all values of k for which A, B, C have equal chances of winning.

Problem 3.4. How many 100-element subsets of the set $\{1, 2, \dots, 2000\}$ have the sum of their elements a multiple of 5?

Problem 3.5 (IMO 1995). Let p be an odd prime number. How many p -element subsets A of $\{1, 2, \dots, 2p\}$ are there, the sum of whose elements is divisible by p ?

Problem 3.6 (Rioplattense Olympiad). Find the number of subsets S of $\{1, 2, \dots, n\}$ which satisfy the following condition: the difference between the number of even elements and odd elements of S is divisible by 3.

Problem 3.7 (Gabriel Dospinescu). Let $f(n)$ be the number of subsets of $\{1, 2, 3, \dots, n\}$ whose elements sum to 0 (mod n). (The empty set is included.) Prove that

$$f(n) = \frac{1}{n} \sum_{\substack{d \mid n \\ d \text{ odd}}} \varphi(d) 2^{n/d}.$$

Problem 3.8 (China TST 2010). Colour each element $x \in M = \{1, 2, \dots, n\}$ either red, blue or yellow. Let

$$A = \{(x, y, z) \in M^3 \mid x, y, z \text{ are the same colour and } n \mid (x + y + z)\}$$

and define

$$B = \{(x, y, z) \in M^3 \mid x, y, z \text{ are distinct colours and } n \mid (x + y + z)\}.$$

Prove that $2|A| \geq |B|$.

Problem 3.9 (USA TSTST 2018). Let $S = \{1, \dots, 100\}$, and for every positive integer n define

$$T_n = \{(a_1, \dots, a_n) \in S^n \mid a_1 + \dots + a_n \equiv 0 \pmod{100}\}.$$

Determine which n have the following property: if we color any 75 elements of S red, then at least half of the n -tuples in T_n have an even number of coordinates with red elements.

Problem 3.10 (IMO Shortlist 1999). Let $p > 3$ be a prime number. For each nonempty subset T of $\{0, 1, 2, 3, \dots, p-1\}$, let $E(T)$ be the set of all $(p-1)$ -tuples $(x_1, \dots, x_{p-1})$, where each $x_i \in T$ and $x_1 + 2x_2 + \dots + (p-1)x_{p-1}$ is divisible by p . Prove that

$$|E(\{0, 1, 3\})| \geq |E(\{0, 1, 2\})|$$

with equality if and only if $p = 5$.

Problem 3.11. Let $a_1, a_2, \dots, a_n$ be nonzero elements of $\mathbb{F}_p$ and $b \in \mathbb{F}_p$. Find the number of solutions (in $\mathbb{F}_p$ to the equation

$$a_1x^2 + a_2x^2 + \dots + a_nx_n^2 = b.$$

Problem 3.12 (China TST 2017). Find the number of ordered arrays $(x_1, \dots, x_{100})$ that satisfies the following conditions:

1. $x_i \in \{1, 2, \dots, 2017\}$ for all i
2. 2017 divides $x_1 + x_2 + \dots + x_{100}$
3. 2017 divides $x_1^2 + x_2^2 + \dots + x_{100}^2$

Problem 3.13 (St Petersburg MO 1991). We call a finite sequence $\{a_k\}_{1 \leq k \leq n}$, p -balanced if the sums

$$s(k, p) = a_k + a_{k+p} + a_{k+2p} + \dots$$

are all equal among themselves for $k = 1, 2, \dots, p$.

Prove that if a sequence of 50 reals is 3, 5, 7, 11, 13 and 17-balanced, then all its terms are equal with 0.

Problem 3.14 (Erdos-Sun). Prove that if a family of k arithmetic progressions covers 2^k consecutive integers, then it covers all of $\mathbb{Z}$.

Problem 3.15 (St. Petersburg MO). Let p be a prime and $n \geq p$ and $a_1, a_2, \dots, a_n$ be integers. Let $f_0 = 1$ and f_k be the number of subsets $I \subset \{1, 2, 3, \dots, n\}$ such that $|I| = k$ and p divides $\sum_{i \in I} a_i$. Prove that p divides

$$\sum_{k=0}^n f_k (-1)^k.$$

Problem 3.16 (Math Reflections). Let p be a prime and n, s positive integers. Prove that p^q divides

$$\sum_{\substack{0 \leq k \leq n \\ p \nmid k-s}} (-1)^k \binom{n}{k}$$

where $q = \left\lfloor \frac{n-1}{p-1} \right\rfloor$

Problem 3.17 (USA TST 2010). Determine whether or not there exists a positive integer k such that $p = 6k + 1$ is a prime and

$$\binom{3k}{k} \equiv 1 \pmod{p}.$$

Solutions

3.1 Let P denote the probability. By the roots of unity filter,

$$P = \frac{1}{5 \cdot 6^n} \sum_{1 \leq x_i \leq 6} \sum_{j=0}^4 \zeta^{j(x_1+x_2+\dots+x_n)} = \frac{1}{5 \cdot 6^n} \sum_{j=0}^4 (\zeta^j + \zeta^{2j} + \dots + \zeta^{6j})^n$$

Because $x \rightarrow jx$ is a permutation on $\mathbb{Z}/5\mathbb{Z}$, it follows that

$$\zeta^j + \zeta^{2j} + \dots + \zeta^{6j} = \zeta^{6j} = \zeta^j$$

for $j \neq 0$. Hence

$$P = \frac{1}{5} + \frac{1}{5 \cdot 6^n} \sum_{j=0}^4 \zeta^{jn} = \begin{cases} \frac{1}{5} - \frac{1}{6^n} & \text{if 5 divides } n, \\ \frac{1}{5} & \text{otherwise.} \end{cases}$$

3.2 Let N denote the number of such tuples and $\zeta = e^{\frac{2\pi i}{p}}$. Applying the roots of unity filter,

$$\begin{aligned} N &= \frac{1}{p} \sum_{x_i \leq n} \sum_{j=0}^{p-1} \zeta^{j(x_1+2x_2+\dots+(p-1)x_{p-1})} \\ &= \frac{1}{p} \sum_{j=0}^{p-1} \prod_{k=1}^{p-1} (\zeta^{jk} + \zeta^{2jk} + \dots + \zeta^{jkn}) \\ &= \frac{n^{p-1}}{p} + \frac{1}{p} \sum_{j=1}^{p-1} \prod_{k=1}^{p-1} \frac{\zeta(\zeta^{jkn} - 1)}{\zeta^{jk} - 1} \end{aligned}$$

If n is divisible by p , then $\frac{\zeta(\zeta^{jkn}-1)}{\zeta^{jk}-1}$ for all j, k . On the other hand, if n is not divisible by p , then $jk \rightarrow jkn$ is a permutation mod p , so

$$\prod_{k=1}^{p-1} \frac{\zeta(\zeta^{jkn} - 1)}{\zeta^{jk} - 1} = 1.$$

Thus

$$N = \begin{cases} \frac{n^{p-1}}{p} & \text{if } p \text{ divides } n. \\ \frac{n^{p-1}+p-1}{p} & \text{if } p \text{ does not divide } n. \end{cases}$$

3.3 Let $\zeta = e^{\frac{2\pi i}{3}}$. Consider the quantity

$$N = \sum_{\substack{T \subseteq S \\ |T|=k}} \zeta^{\sum_{t \in T} t} = c_0 + c_1 \zeta + c_2 \zeta^2.$$

The three are equal if and only if $N = 0$. (This is a corollary of the fact that $1 + X + X^2$ is the minimal polynomial of ζ . See the solution to the next problem for a more complete explanation.) However, N is precisely the coefficient of X^k in the polynomial

$$(X + \zeta)(X + \zeta^2) \dots (X + \zeta^{1986}) = (X^3 + 1)^{662}.$$

Thus the answer is the integers between 1 and 1986 not divisible by 3.

3.4 Let $\zeta = e^{\frac{2\pi i}{5}}$ and N denote the number we are seeking. By the roots of unity filter,

$$N = \frac{1}{5} \sum_{j=0}^4 \sum_{\substack{S \subseteq [2000] \\ |S|=100}} \zeta^{j \sum_{s \in S} s}$$

If $j = 0$, the inner sum evaluates to $\binom{2000}{100}$. Otherwise, it is precisely the coefficient of X^{100} in the polynomial

$$(X + \zeta^j)(X + \zeta^{2j}) \dots (X + \zeta^{2000j}) = (X^5 + 1)^{400}.$$

By the binomial theorem, this is $\binom{400}{20}$. Hence

$$N = \frac{1}{5} \left(\binom{2000}{100} + 4 \binom{400}{20} \right).$$

3.5 Let $\zeta = e^{\frac{2\pi i}{p}}$, and consider the quantity

$$N = \sum_{\substack{S \subseteq [2p] \\ |S|=p}} \zeta^{\sum_{s \in S} s}.$$

On one hand

$$N = c_0 + c_1 \zeta + \dots + c_{p-1} \zeta^{p-1},$$

where the numbers c_i represent the number of subsets S of $\{1, 2, \dots, 2p\}$ with size p that have a sum $i \pmod{p}$. On the other hand, N is the coefficient of X^p in

$$(X + \zeta)(X + \zeta^2) \dots (X + \zeta^{2p}) = (X^p + 1)^2,$$

which is evidently 2. Thus

$$(c_0 - 2) + c_1 \zeta + c_{p-1} \zeta^{p-1} = 0$$

This implies that

$$1 + X + X^2 + \dots + X^{p-1} \text{ divides } (c_0 - 2) + c_1 X + c_2 X^2 + \dots + c_{p-1} X^{p-1}$$

so $c_0 - 2 = c_1 = \dots = c_{p-1}$. Since $c_0 + c_1 + \dots + c_{p-1} = \binom{2p}{p}$, it follows that

$$c_0 = \frac{\binom{2p}{p} - 2}{p}.$$

3.6 Let ζ be a primitive third root of unity and N be the number we are seeking. Suppose first that n is even and write $n = 2m$. Then

$$N = \sum_{k=0}^m \binom{m}{k} \sum_{\substack{k \equiv \ell \\ (\text{mod } 3)}} \binom{k}{\ell}.$$

Thus

$$\begin{aligned} N &= \frac{1}{3} \sum_{k=0}^m \binom{m}{k} \sum_{\ell=0}^m \binom{m}{\ell} \sum_{j=0}^2 \zeta^{j(k-\ell)} \\ &= \frac{2^n}{3} + \sum_{j=1}^2 \left(\sum_{k=0}^m \binom{m}{k} \zeta^{jk} \right) \left(\sum_{\ell=0}^m \binom{m}{\ell} \zeta^{-j\ell} \right) \\ &= \frac{1}{3} \sum_{j=0}^2 (1 + \zeta^j)^m (1 + \zeta^{-j})^m \\ &= \frac{2^n}{3} + \frac{2}{3} (1 + \zeta)^m (1 + \zeta^2)^m \\ &= \frac{2^n + 2}{3} \end{aligned}$$

The case where n is odd is left as an exercise.

3.7 Let $\zeta = e^{\frac{2\pi i}{p}}$. Applying the roots of unity filter,

$$f(n) = \frac{1}{n} \sum_{j=0}^{p-1} \prod_{k=1}^n (1 + \zeta^{jk})$$

Suppose that ζ^j is a root of unity of order d . $(1, \zeta^{jk}, \dots, \zeta^{n(p-1)})$ counts each of the roots of unity $(1, \zeta, \dots, \zeta^{d-1})$ exactly n/d times. Thus

$$f(n) = \frac{1}{n} \sum_{d|n} \varphi(d) \left(\prod_{k=1}^{d-1} (1 + \zeta^k) \right)^{n/d}.$$

Then it suffices to show that

$$\prod_{k=1}^{d-1} (1 + \zeta^k) = \begin{cases} 2 & \text{if } d \text{ is odd,} \\ 0 & \text{otherwise.} \end{cases}$$

Let $g(X) = \prod_{k=1}^{d-1} (X - \zeta^k) = \frac{X^d - 1}{X - 1}$. Hence

$$\prod_{k=1}^{d-1} (1 + \zeta^k) = (-1)^{d-1} g(-1) = \begin{cases} 2 & \text{if } d \text{ is odd,} \\ 0 & \text{otherwise.} \end{cases}$$

This completes the proof.

3.8 Let $\zeta = e^{\frac{2\pi i}{n}}$ and denote

$$R_j = \sum_{r \text{ red}} \zeta^{jr}$$

Define B_j and Y_j similarly. The roots of unity filter gives

$$|A| = \frac{1}{n} \sum_{j=0}^{p-1} R_j^3 + B_j^3 + C_j^3 \text{ and } |B| = \frac{6}{n} \sum_{j=0}^{p-1} R_j B_j Y_j$$

Therefore

$$\begin{aligned} 2|A| - |B| &= \frac{2}{p} \sum_{j=0}^{p-1} R_j^3 + B_j^3 + Y_j^3 - 3R_j B_j C_j \\ &= \frac{2}{p} \sum_{j=0}^{p-1} (R_j + B_j + Y_j)(R_j^2 + B_j^2 + Y_j^2 - B_j Y_j - Y_j R_j - R_j B_j) \end{aligned}$$

If $j \neq 0$, then $R_j + B_j + Y_j = 0$. Thus it suffices to show that

$$n(R_0^2 + B_0^2 + Y_0^2 - B_0 Y_0 - Y_0 R_0 - R_0 B_0) \geq 0$$

which is true by AM-GM.

3.9 The answer is all even n . Colour all the uncoloured elements of S blue. Let $\zeta = e^{\frac{2\pi i}{100}}$ and define

$$R_j = \sum_{r \text{ red}} \zeta^{jr} \text{ and } B_j = \sum_{b \text{ blue}} \zeta^{jb}.$$

Let M and N denote the number of tuples in T_n with even and odd coordinates, respectively. By the roots of unity filter, the number of tuples with precisely k red elements is given by

$$\frac{1}{100} \binom{n}{k} \sum_{j=0}^{100} R_j^k B_j^{n-k}.$$

Thus

$$\begin{aligned} M - N &= \frac{1}{100} \sum_{j=0}^{99} \left(\sum_{k \text{ even}} \binom{n}{k} R_j^k B_k^{n-k} - \sum_{k \text{ odd}} \binom{n}{k} R_j^k B_j^{n-k} \right) \\ &= \frac{1}{100} \sum_{j=0}^{99} (B_j - R_j)^n \end{aligned}$$

For $j \neq 0$, it is easy to see that $B_j + R_j = 0$. Thus

$$\begin{aligned} M - N &= \frac{1}{100} \left((B_0 - R_0)^n + \sum_{j=1}^{99} (2B_j)^n \right) \\ &= \frac{1}{100} \left((-50)^n - 50^n + 2^n \sum_{j=0}^{99} B_j^n \right) \end{aligned}$$

The quantity

$$\frac{1}{100} \sum_{j=0}^{99} B_j^n \tag{†}$$

counts the number of tuples in T_n with only blue coordinates, so it is non-negative. Hence if n is even, the quantity $M - N$ is non-negative, and we are done.

On the other hand, consider colouring all $b \equiv 2 \pmod{4}$ blue for odd n . Then the quantity in (†) is zero, so $M - N = -50^{n-1} < 0$.

3.10 Let $\zeta = e^{\frac{2\pi i}{p}}$. Applying the roots of unity filter,

$$\begin{aligned} |E(\{0, 1, 2\})| &= \frac{1}{p} \sum_{x_i \in \{0, 1, 2\}} \sum_{j=0}^{p-1} \zeta^{j(x_1 + 2x_2 + \dots + (p-1)x_{p-1})} \\ &= \frac{1}{p} \sum_{j=0}^{p-1} \prod_{i=1}^{p-1} (1 + \zeta^{ij} + \zeta^{2ij}) \\ &= \frac{3^{p-1}}{p} + \sum_{j=1}^{p-1} \prod_{i=1}^{p-1} \frac{\zeta^{3ij} - 1}{\zeta^{ij} - 1} \\ &= \frac{3^{p-1} + (p-1)}{p}. \end{aligned}$$

Likewise, if α, β, γ are the roots of the polynomial $X^3 + X + 1$, then

$$\begin{aligned} |E(\{0, 1, 3\})| &= \frac{3^{p-1}}{p} + \frac{1}{p} \sum_{j=1}^{p-1} \prod_{i=1}^{p-1} (1 + \zeta^{ij} + \zeta^{3ij}) \\ &= \frac{3^{p-1}}{p} + \frac{1}{p} \sum_{j=1}^{p-1} \prod_{i=1}^{p-1} (\zeta^i - \alpha)(\zeta^i - \beta)(\zeta^i - \gamma) \\ &= \frac{3^{p-1}}{p} + \frac{p-1}{p} \cdot \frac{(\alpha^p - 1)(\beta^p - 1)(\gamma^p - 1)}{(\alpha - 1)(\beta - 1)(\gamma - 1)}. \end{aligned}$$

Hence it suffices to prove $a_p \geq 1$ with equality if and only if $p = 5$ where $a_n = \frac{(\alpha^n - 1)(\beta^n - 1)(\gamma^n - 1)}{(\alpha - 1)(\beta - 1)(\gamma - 1)}$. Because a_n is a linear combination of $\alpha^n, \beta^n, \gamma^n, (\alpha\beta)^n, (\beta\gamma)^n, (\gamma\alpha)^n$, and $(\alpha\beta\gamma)^n$, it follows that a_n satisfies a recursion with the characteristic polynomial

$$\begin{aligned} &(X - \alpha)(X - \beta)(X - \gamma)(X - \alpha\beta)(X - \beta\gamma)(X - \gamma\alpha)(X - \alpha\beta\gamma) \\ &= (X^3 + X + 1)(X^3 - X^2 - 1)(X + 1) \\ &= x^7 - 2x^3 - 2x^2 - 2x - 1 \end{aligned}$$

Thus there exists a constant C such that the sequence a_n obeys the recursive relation

$$a_{n+7} = 2a_{n+3} + 2a_{n+2} + 2a_{n+1} + a_n + C$$

To determine C , calculate the first 8 values of a_n :

n	0	1	2	3	4	5	6	7
a_n	0	1	1	3	1	1	3	8

This implies $C = -2$. From here, it is easy to see that a_n is increasing after a_6 , so we are done.

3.11 The answer is

$$\begin{cases} p^{n-1} - (-1)^{\frac{n(p-1)}{4}} \left(\frac{a_1 a_2 \dots a_n}{p} \right) p^{\frac{n}{2}-1} & \text{if } n \text{ is even} \\ p^{n-1} - (-1)^{\frac{(n+1)(p-1)}{4}} \left(\frac{-ba_1 a_2 \dots a_n}{p} \right) p^{\frac{n-1}{2}} & \text{if } n \text{ is odd.} \end{cases}$$

The solution is almost identical to [example 2.1](#).

3.12 The answer is 2017^{98} . Let $p = 2017$ and N denote the number of such ordered arrays. By the roots of unity filter,

$$\begin{aligned} p^2 N &= \sum_{(x_1, \dots, x_{100})} \sum_{a=0}^{p-1} \zeta^{a(x_1^2 + x_2^2 + \dots + x_{100}^2)} \sum_{b=0}^{p-1} \zeta^{b(x_1 + x_2 + \dots + x_{100})} \\ &= \sum_{a=0}^{p-1} \sum_{b=0}^{p-1} \left(\sum_{x=0}^{p-1} \zeta^{ax^2 + bx} \right)^{100} \end{aligned}$$

Before proceeding, we take a moment to deal with the case $a = 0$. Since $1 + \zeta^b + \dots + \zeta^{b(p-1)} = 0$ if $b \neq 0$, we have

$$\sum_{b=0}^{p-1} \left(\sum_{x=0}^{p-1} \zeta^{bx} \right)^{100} = p^{100}.$$

In order to apply our knowledge about Gauss sums, we complete the square:

$$pN^2 = p^{100} + \sum_{a=1}^{p-1} \sum_{b=0}^{p-1} \zeta^{-\frac{25b^2}{a}} \left(\sum_{x=0}^{p-1} \zeta^{a(x + \frac{b}{2a})^2} \right)^{100}.$$

Since $x \rightarrow x + \frac{b}{2a}$ is a permutation, it follows that

$$\begin{aligned} p^2 N &= p^{100} + \sum_{a=1}^{p-1} \sum_{b=0}^{p-1} \zeta^{-\frac{25b^2}{a}} G(a)^{100} \\ &= p^{100} + p^{50} \sum_{a=1}^{p-1} \sum_{b=0}^{p-1} \zeta^{-\frac{25b^2}{a}} \\ &= p^{100} + p^{50} + \sum_{a=1}^{p-1} a = 1^{p-1} G\left(-\frac{25}{a}\right) \\ &= p^{100} + p^{50} \sum_{a=1}^{p-1} \left(\frac{-25/a}{p} \right) G \\ &= p^{100} \end{aligned}$$

The final step comes from the fact that $a \rightarrow -25/a$ is bijective, so

$$\sum_{a=1}^{p-1} \left(\frac{-25/a}{p} \right) G = \sum_{a=1}^{p-1} \left(\frac{a}{p} \right) G = G \cdot \frac{p-1}{2} - G \cdot \frac{p-1}{2}.$$

3.13 For the moment, fix a prime p and pick an arbitrary primitive p -th root of unity ζ . Consider the function

$$f(X) = a_1X + a_2X^2 + \cdots + a_{50}X^{50}.$$

Because $S(1, p) = S(2, p) = \cdots = S(p, p)$, it follows that

$$f(\zeta) = \sum_{k=1}^p S(k, p)\zeta^k = 0.$$

There are $2 + 4 + 6 + 10 + 12 + 16 = 50$ roots of unity with order 3, 5, 7, 11, 13, or 17, so it follows that f has 51 roots (0 as well). Thus $f(X) = 0$, and so the a_i are all zero.

3.14 Denote the arithmetic progressions by $\{a_i + td_i\}_{t \in \mathbb{Z}}$. Let $\zeta_j = e^{\frac{2\pi i}{d_j}}$. An integer n belongs to one of the arithmetic sequences if and only if

$$\prod_{j=1}^k (1 - \zeta_j^{m-a_j}) = 0.$$

On the other hand,

$$\prod_{j=1}^k (1 - \zeta_j^{m-a_j}) = \sum_{S \subseteq [k]} (-1)^{|S|} \prod_{j \in S} \zeta_j^{m-a_j} = \sum_{S \subseteq [k]} \alpha_S z_S^m,$$

where $\alpha_S = (-1)^S \prod_{j \in S} \zeta_j^{-a_j}$ and $z_S = \prod_{j \in S} \zeta_j$. Let $u_n = \sum_{S \subseteq [k]} \alpha_S z_S^n$, so n belongs to one of the arithmetic progressions if and only if $u_n = 0$. The u_n satisfy a linear recurrence of order 2^k ; for proof, consider the polynomial

$$f(X) = \prod_{S \subseteq [k]} (X - z_S) = c_0 + c_1X + \cdots + c_{2^k}X^{2^k}.$$

Because the arithmetic progression covers 2^k consecutive integers, it follows that 2^k consecutive terms u_n are zero, so $u_n = 0$ for all $n \in \mathbb{Z}$.

3.15 Applying the roots of unity filter (verify the equality when $k = 0$):

$$\begin{aligned} \sum_{k=0}^n f_k (-1)^k &= \sum_{k=0}^n \frac{(-1)^k}{p} \sum_{\substack{I \subseteq [n] \\ |I|=k}} \sum_{j=0}^{p-1} \zeta^{j \sum_{i \in I} a_i} \\ &= \sum_{k=0}^n \left(\frac{(-1)^k}{p} \binom{n}{k} + \sum_{j=1}^{p-1} \sum_{\substack{I \subseteq [n] \\ |I|=k}} \zeta^{j \sum_{i \in I} a_i} \right) \\ &= \frac{1}{p} \sum_{j=1}^{p-1} \sum_{I \subseteq [n]} (-1)^{|I|} \zeta^{j \sum_{i \in I} a_i} \\ &= \frac{1}{p} \sum_{j=1}^{p-1} \prod_{i=1}^n (1 - \zeta^{ja_i}) \end{aligned}$$

Thus it suffices to show that

$$p^2 \mid \sum_{j=1}^{p-1} \prod_{i=1}^n (1 - \zeta^{ja_i})$$

Denote this quantity by N . To prove this divisibility, we invoke some p -adic theory. Let $S = \{f(\zeta) \mid f \in \mathbb{Z}[x]\}$ and $n \cdot S = \{nz \mid z \in S\}$ for any integer n . First we establish a useful lemma:

Claim. Let N be a complex number. If $N \in n \cdot S$, then N is divisible by n (in $\mathbb{Z}$).

Proof. By definition, there exists an integer polynomial f such that $nf(\zeta) = N$. Since the minimal polynomial of ζ has degree $p-1$, there are integers $a_0, a_1, \dots, a_{p-2}$ such that

$$N = n(a_0 + a_1\zeta + \dots + a_{p-1}\zeta^{p-1}).$$

However, this implies

$$N - na_0 = na_1 = na_2 = \dots = na_{p-1} \implies n \mid N$$

□

Returning to the problem at hand, rewrite

$$N = (1 - \zeta)^n \sum_{j=1}^{p-1} \prod_{i=1}^n (1 + \zeta + \dots + \zeta^{ja_i-1})$$

Let $1 - \zeta = \xi$. We claim that $\xi^{p-1} \in p \cdot S$. (Think of this as saying $\nu_p(\xi) = \frac{1}{p-1}$ when the notion of ν_p is extended to a ring of algebraic integers $\mathcal{O}_k$ over the field $\mathbb{Q}(\zeta)$. For more information, research p -adic numbers. Here we use just elementary methods.) Observe that

$$\zeta^p = 1 = (1 - \xi)^p = \sum_{k=0}^p \binom{p}{k} (-\xi)^k.$$

It follows that

$$\xi^{p-1} = \sum_{k=1}^{p-1} \binom{p}{k} (1 - \zeta^{k-1}) \in p \cdot S.$$

Then $N \in p^n \cdot S$. Moreover, $N^{p-1} \in p^{(p-1)n}$. Since $n \geq p$, it follows that N^{p-1} is divisible by p^n , which implies N is divisible by p^2 .

3.16 The solution proceeds very similarly to the previous problem. Let $\zeta = e^{\frac{2\pi i}{p}}$. By the roots of unity filter,

$$\sum_{\substack{0 \leq k \leq n \\ p \mid k-s}} (-1)^k \binom{n}{k} = \frac{1}{p} \sum_{j=1}^{p-1} \sum_{k=0}^n (1 - \zeta^{j(k-s)})^n$$

When $j = 0$, every term is 0. As in the previous problem, let $S = \{f(\zeta) \mid f \in \mathbb{Z}[x]\}$ and $n \cdot S = \{ns \mid s \in S\}$. Then $(1 - \zeta)^{p-1} \in p \cdot S$. It suffices to show that

$$p^{q+1} \mid \sum_{j=0}^{p-1} \sum_{k=0}^n (1 - \zeta^{j(k-s)})^n$$

Denote this quantity, by N . Then

$$N = (1 - \zeta^n) \sum_{j=0}^{p-1} \sum_{k=0}^n (1 + \zeta + \dots + \zeta^{j(k-s-1)})^n$$

Since $n > q(p-1)$, it follows that $N^{p-1} \in p^{q(p-1)+1}S$, so p^{q+1} divides N , as desired.

3.17 This solution does not concretely apply the roots of unity filter (no complex numbers are used), but heavily employs its spirit. It is meant to demonstrate that the roots of unity filter is only the tip of a much larger mathematical iceberg. An interested reader is encouraged to research finite Fourier analysis. For now, here is the solution to the problem.

The answer is negative. All work is done in $\mathbb{Z}/p\mathbb{Z}$.

Let ζ be an element of $\mathbb{Z}/p\mathbb{Z}$ with order k (i.e. ζ is a k -th root of unity). Then

$$\frac{1}{k} \sum_{j=0}^{k-1} (1 + \zeta^j)^{3k} = \frac{1}{k} \sum_{i=0}^{3k} \binom{3k}{i} \sum_{j=0}^{k-1} \zeta^{ji} = \binom{3k}{0} + \binom{3k}{k} + \binom{3k}{2k} + \binom{3k}{3k}.$$

Suppose for the sake of contradiction that $\binom{3k}{k} = 1$. Then

$$\sum_{j=0}^{k-1} (1 + \zeta^j)^{3k} = 4k.$$

However, $(1 + \zeta^j)^{3k} \in \{-1, 1\}$, so this is impossible.